AMRITESH PRAVEEN

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OVERVIEW

AI/ML Engineer focused on building retrieval-augmented systems, intelligent agents, and production AI applications. Experienced in end-to-end development spanning data pipelines, model integration, backend services, and deployment.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript
AI/ML: PyTorch, TensorFlow, Scikit-learn, NLP, Deep Learning, Computer Vision
Generative AI: RAG, Hybrid Retrieval, Embeddings, Reranking, LangChain
Backend & Data: FastAPI, REST APIs, Pandas, NumPy, ChromaDB, pgvector
Frameworks & Infrastructure: Astro, Cloudflare, GCP, Docker, Vercel, Git, GitHub

EXPERIENCE

AI & ML Associate - Impact Solutions Lab **Jun 2026 - Present**

- Designed a multilingual Survey Intelligence system using RAG and Text-to-SQL, enabling natural-language querying of community survey data.
- Built a student-specific RAG assistant over class recordings and learning materials, where each student's answers draw only on their own classes.
- Created an internal examination platform enabling teachers to create, administer, and evaluate school-level assessments.
- Rebuilt Adira Academy's public website, replacing WordPress with Astro, a Git-backed CMS, and a zero-hosting-cost deployment.
- Developed Adira Academy's admissions platform, replacing manual workflows with digital registration, application tracking, examination scheduling, and staff management using FastAPI and Astro.

AI & ML Intern - Impact Solutions Lab **Oct 2025 - May 2026**

- Built AdiraExams, an entrance examination platform replacing third-party software and serving 300+ students.
- Automated Google Drive-to-YouTube publishing and Zoom attendance reporting using Python, Google APIs, and scheduled cloud workflows.

Machine Learning Intern - SixPhrase **Feb 2025 - May 2025**

- Built supervised classification models reaching 87% accuracy, and cut training time by about 25% with PCA and LDA dimensionality reduction.

Data Analysis Intern - Miles Education **Apr 2024 - May 2024**

- Analyzed datasets using Python, Pandas, and NumPy to identify trends and support data-driven decisions.

PROJECTS

RAG Lab - Controlled Comparison of RAG Architectures GitHub

- Implemented eight RAG architectures, including HyDE, Adaptive, Corrective, Self-Reflective, and Agentic RAG, using a shared retrieval and generation framework.
- Built a reproducible evaluation pipeline using a frozen 120-question dataset, digest-verified inputs, provenance checks, and blind LLM-based judging; evaluation is in progress across all eight architectures.

Hunter Wumpus - Reinforcement Learning Agent Portfolio

- Developed a PPO-based reinforcement learning agent that hunts the player through fog of war in a dynamic Hunter Wumpus environment; the subject of a published paper.

PUBLICATIONS & PATENTS

- **Reinforcement Learning for Constraint Satisfaction Game Agents: From Tabular Methods to Adversarial Deep Policy Optimisation.** Amritesh Praveen and Gagan Venkat, May 2026. DOI
- **Railway Track Monitoring and Repairing Device.** Indian Patent Application No. 202541017469 A, March 2025.

EDUCATION

Jain University **2022 - 2026**
B.Tech in Computer Science Engineering- Artificial Intelligence and Machine Learning CGPA: 7.3/10

CERTIFICATIONS

- Machine Learning Specialization - Stanford University and DeepLearning.AI
- Natural Language Processing with Attention Models - DeepLearning.AI
- Fundamentals of Reinforcement Learning - University of Alberta